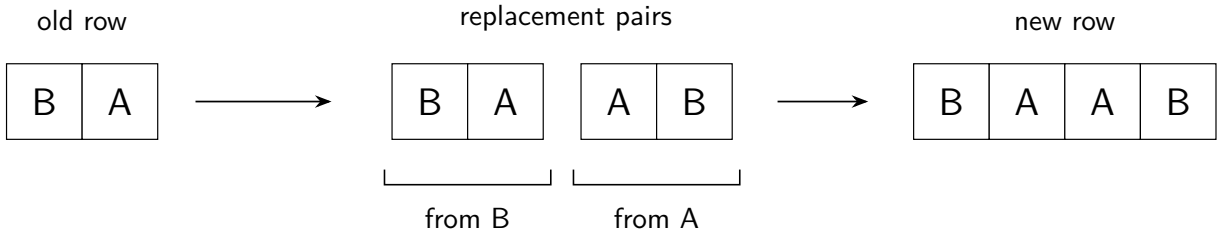
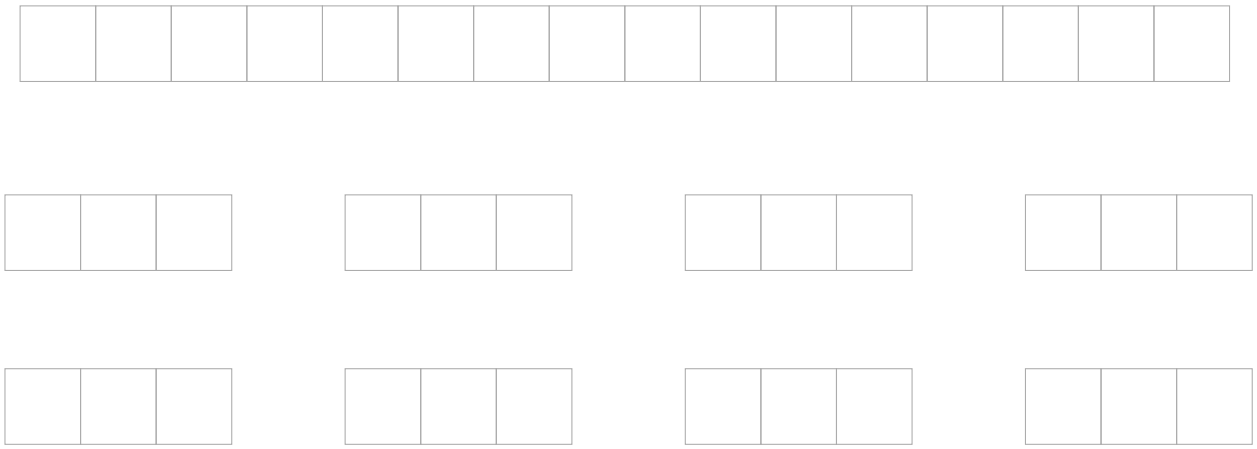
Replace A by AB and B by BA. Replace each old tile once. Keep the old and new rows separate while your partner checks.

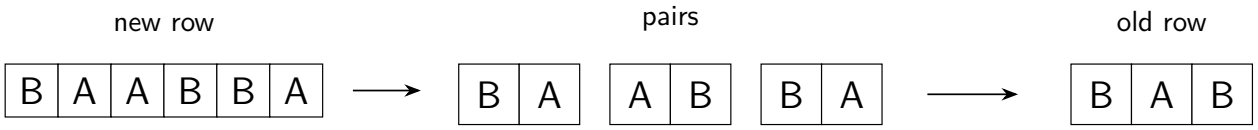


Problem 1: Grow a row of 16 tiles from A. Find every different strip of three neighboring tiles in your row. Record each short strip once.



Problem 2: Could AAA or BBB ever appear in a longer row grown from A? Explain why, or build a row that contains one.

Problem 3: Undo a replacement by turning each pair AB into A and each pair BA into B.



Each row below claims to be a whole row grown from a single A. Which claims can be right? Show what settles each claim.

A	B	B	A	B	A	A	B

A	B	B	A	A	B	B	A

A	B	B	A	B	B	A	B

A	B	B	A	B	A	A	B	A	B	B	A	B	A	A	B